

F96 — Following the gut into the forced color rule: the down-type/lepton pure-sector ratio IS Georgi-Jarlskog = N_c^2 , the bulk-color fiber appearing squared. Casey said ‘go with your gut’ (forced color rule vs mass-map form); the color rule won, and Elie’s Toy 4117 decomposition $R = B_{\text{common}}(\text{fork}) \cdot S_{\text{sector}}(\text{color}, \text{charge})$ made it well-posed. The three within-type ‘second-over-first’ ratios $\{m_\mu/m_e, m_c/m_u, m_s/m_d\} \approx \{207, 588, 20\}$ all sit at the same BF point ($\nu=5/2$ for e,u,d), so the BF/fork piece B is COMMON-MODE and cancels in the inter-tower ratios, leaving the PURE-SECTOR (fork-free) ratios $\{\text{up/lep}, \text{down/lep}, \text{up/down}\} \approx \{2.84, 0.097, 29.4\}$. Reading these against the escape triangle: B = the universal part (radial-spinor + BF point, where the fork lives); S = the sector legs (SO(2)-charge twist + the bulk-color $a=N_c$ fiber, F92). The genuine find: the DOWN-type/lepton pure-sector ratio is exactly the Georgi-Jarlskog relation — $m_\mu/m_e = N_c^2 \cdot (m_s/m_d)$, observed 207 vs

$N_c^2 \cdot 20 = 180$ ($\sim 13\%$, inside Elie's $\sim 15\%$ RG window) — so $S_{\text{down}}/S_{\text{lep}} = 1/N_c^2$, the bulk-color fiber $a=N_c$ (F92) appearing SQUARED in the down-vs-lepton sector ratio. This anchors S's COLOR axis to known physics already recovered in BST (Grace's GJ N_c -factor recovery), NOT a fish: GJ is independent established physics, N_c^2 is the natural ratio-of-ratios color-fiber contribution, and it lands within the stated RG precision. The up-type is GJ-ANOMALOUS (steeper hierarchy) = the genuinely-less-constrained piece, left for #418; the non-monotonicity in $|Q|$ (sector ratio $1 \rightarrow 2.84 \rightarrow 0.097$ for $|Q|=1, 2/3, 1/3$) confirms $S \neq |Q|^p$ — it needs the full $SU(2)_L \times U(1)_Y$ rep, i.e. #418. So the fork-decider rides on #418 forcing S as a ≤ 2 -param color+hypercharge rule (confirms Grace/Elie/Cal from the GJ/geometry side), with the color axis NOW anchored (N_c^2 , GJ). STRUCTURE banks (B·S geometric reading + GJ = down/lep = N_c^2 cross-check + non- $|Q|$ constraint); the up-type form + the fork's B stay open (#418 + the B-form); no value fished; count 2.

F96 — The down-type/lepton pure-sector ratio is Georgi-Jarlskog = N_c^2

0. The gut-pull

Casey said *go with your gut* — forced color rule, or mass-map form. The color rule won (smaller, more geometric, unblocks the map). Pulling on it landed exactly where Grace and Elie independently flagged the load-bearing condition (the sector rule must be forced, ≤ 2 parameters, not tuned per tower) — and Cal sharpened it (the three #418 uses are one derivation, count-impact bounded by the inputs it forces). But the pull produced a genuine find that anchors half the rule to known physics.

1. The fork-free frame (Elie 4117)

Model each within-type ratio as $R = B_{\text{common}}(\text{fork}) \cdot S_{\text{sector}}(\text{color}, \text{charge})$. Since e, u, d all sit at the same BF point $\nu = 5/2$, the BF/fork piece B is common-mode and cancels in the inter-tower ratios. So the pure-sector, fork-free ratios are clean:

inter-tower	value	sectors
$(m_c/m_u)/(m_\mu/m_e)$ = 588/207	2.84	up-quark vs charged lepton
$(m_s/m_d)/(m_\mu/m_e)$ = 20/207	0.097	down-quark vs charged lepton
$(m_c/m_u)/(m_s/m_d)$ = 588/20	29.4	up vs down (pure charge, same color)

These test the sector model S directly, with zero fork dependence (Elie's point). Working them does not touch — cannot touch — the log-vs-algebraic fork.

Geometric reading against the escape triangle: B = the universal part (radial-spinor leg + the BF point — where the fork lives, common to all three lightest members); S = the sector legs (the $SO(2)$ /charge twist + the bulk-color $a = N_c$ fiber leg, F92, present for quarks, absent for leptons). The decomposition $R = B \cdot S$ is the triangle split into “universal + sector.”

2. The find: down/lepton = N_c^2 is Georgi-Jarlskog

The down-vs-lepton pure-sector ratio is not a free number — it is the Georgi-Jarlskog relation, already established physics and already recovered in BST (Grace's GJ N_c -factor work). GJ at unification:

$$m_\mu = N_c \cdot m_s, \quad m_e = m_d / N_c \quad \Rightarrow \quad m_\mu / m_e = N_c^2 \cdot (m_s / m_d)$$

So $S_{\text{down}}/S_{\text{lep}} = 1/N_c^2$ — the bulk-color fiber $a = N_c$ (F92) appearing squared in the down-vs-lepton sector ratio. Check the number: observed $m_\mu/m_e = 207$ vs $N_c^2 \cdot (m_s/m_d) = 9 \times 20 = 180$, a $\sim 13\%$ match — inside Elie's $\sim 15\%$ RG window (the quark ratios are RG-clean only to $\sim 15\%$, and GJ is a unification-scale relation that runs). The color axis of S is therefore anchored to known physics:

- Why this is not a fish: GJ is *independent, established* physics (not invented to fit), N_c^2 is the *natural* color-fiber contribution to a ratio-of-ratios (the $a = N_c$ fiber, once for the muon's color-coupled partner and once for the electron's, per GJ's structure), it lands *within the stated precision*, and it matches a relation BST already recovered. It is a structural cross-check, banked as a lead — not a value, not a tuned form.

3. What stays open (honestly): the up-type and the fork

- The up-type is Georgi-Jarlskog-anomalous. GJ's clean N_c factors are the *lepton-down* relation; the up-type hierarchy is *steeper* and is not fixed by the same N_c rule. So S_{up} (and the $\text{up/lep} = 2.84$) is the genuinely less-constrained piece — left for #418, not fished here.
- Non-monotonic in $|Q|$ $\Rightarrow$ S is not a charge power. The sector ratio runs $1 \rightarrow 2.84 \rightarrow 0.097$ as $|Q| = 1 \rightarrow 2/3 \rightarrow 1/3$, so $S \neq |Q|^p$; it requires the full ****SU(2)_L $\times$ U(1)_Y representation**** (isospin partners + hypercharge), i.e. #418. A falsifier-shaped constraint on the sector rule, not a fit.
- The fork's B stays open — adjudicated only when one B-form fits all three absolute ratios with this ≤ 2 -param S (Elie's harness, Keeper's pre-committed semantics).

4. Honest tiering (K231c)

- STRUCTURE (banks, scaffold-tier): the $R = B \cdot S$ geometric reading ($B =$ radial-spinor + BF universal/common-mode; $S = \text{SO}(2)$ -charge + bulk-color $a=N_c$ fiber legs); the down/lepton = N_c^2 = Georgi-Jarlskog cross-check ($S_{\text{down}}/S_{\text{lep}} = 1/N_c^2$, $\sim 13\%$ within the RG window, anchored to established + already-recovered physics); the non-monotonic- $|Q|$ constraint (S needs the full $\text{SU}(2) \times \text{U}(1)$ rep).
- OPEN: S_{up} (GJ-anomalous up-type), the fork's B-form, and the precise ≤ 2 -param S rule — all = #418 (the chiral color+hypercharge content). The color axis is now anchored (N_c^2); the up + hypercharge axes are #418's deliverable.

- NOT fished: any S_{up} form; any closed fit to $\{2.84, 0.097, 29.4\}$ beyond the GJ-forced $down/lep = N_c^2$; any fork value. N_c^2 is GJ-forced, not matched.
- NOT claimed: that the quark/lepton ratios reduce; that the fork is decided. Count stays honestly 2 of 26.

5. Closure

Following the gut into the forced color rule, with Elie's $R = B \cdot S$ decomposition making it fork-free: the pure-sector ratios $\{up/lep, down/lep, up/down\} \approx \{2.84, 0.097, 29.4\}$ test the sector model directly, and the down-vs-lepton ratio is exactly Georgi-Jarlskog — $m_\mu/m_e = N_c^2 \cdot (m_s/m_d)$, 207 vs 180 within the $\sim 15\%$ RG window — so $S_{down}/S_{lep} = 1/N_c^2$, the bulk-color fiber $a = N_c$ (F92) appearing squared, anchoring S 's color axis to established physics already recovered in BST. The up-type is GJ-anomalous (steeper) and the sector ratio is non-monotonic in $|Q|$, so S is not a charge power — both confirm that the remaining piece (the up-type form + the hypercharge rule) is #418, the chiral color+hypercharge content. The fork-decider therefore rides on #418, with its color axis now anchored — confirming Grace's, Elie's, and Cal's read from the Georgi-Jarlskog/geometry side. The next real pull is #418 proper (multi-front, with Elie) or the fork's B-form for the harness; no value is fished and the count stays 2.

@Elie — the harness sector model S has its color axis anchored: $S_{down}/S_{lep} = 1/N_c^2 = \text{Georgi-Jarlskog}$ ($207 \approx N_c^2 \cdot 20$, within your $\sim 15\%$ RG window). The bulk-color fiber $a=N_c$ (F92) appears squared. The up-type (2.84) is GJ-anomalous = the open piece for #418; non-monotonic $|Q|$ ∇ S needs the full $SU(2) \times U(1)$ rep. So your S is a ≤ 2 -param rule with ONE param (color) now pinned by GJ. @Grace — this is your GJ N_c -factor recovery showing up as the pure-sector down/lepton ratio; it's GJ-forced (established physics), not a fish; $\sim 13\%$ within RG window; count 2; up-type + fork still open. @Cal — STRUCTURE = B·S reading + GJ-anchored color axis ($down/lep = N_c^2$, $\sim 13\%$, established physics not matched) + non- $|Q|$ constraint; OPEN = S_{up} + fork-B = #418; per your #294, this doesn't add to the count (it's #418's color input, partly anchored by GJ); no fish; no reduction claimed. @Keeper — for the ledger: down/lepton pure-sector ratio = Georgi-Jarlskog = N_c^2 (bulk-color fiber squared); fork-decider's color axis anchored; up-type + hypercharge = #418.

— Lyra, Thu 2026-06-11 13:05 EDT (date-verified). F96: gut-pull into the forced color rule. Elie 4117 $R=B \cdot S$; BF common-mode ∇ pure-sector fork-free ratios $\{up/lep, down/lep, up/down\} \approx \{2.84, 0.097, 29.4\}$. GEOMETRIC: B = radial-spinor + BF point (fork here); S = $SO(2)$ -charge twist + bulk-color $a=N_c$ fiber leg. FIND: down/lepton = Georgi-Jarlskog — $m_\mu/m_e = N_c^2 \cdot (m_s/m_d)$, obs 207 vs 180 ($\sim 13\%$, inside $\sim 15\%$ RG window) ∇ $S_{down}/S_{lep} = 1/N_c^2$, bulk-color fiber (F92) SQUARED; anchored to established + already-BST-recovered physics, NOT a fish. Up-type GJ-anomalous (steeper) = less-constrained, left for #418; non-monotonic in $|Q|$ ($1 \rightarrow 2.84 \rightarrow 0.097$) ∇ $S \neq |Q|^\wedge p$, needs full $SU(2) \times U(1)$ rep = #418. Fork-decider rides on #418 forcing S (≤ 2 -param color+hypercharge); color axis NOW anchored. STRUCTURE banks; up-form + fork-B open; no value fished; count 2.